MARIAN SIMARRO GONZALEZ

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https://mariansimarro.github.io/

EDUCATION

2022-2023

MSC COGNITIVE NEUROSCIENCE OF LANGUAGE
Basque Center on Cognition Brain and Language (EHU/UPV), Spain
Thesis: A complementary learning systems approach to computational modelling of bilingual language learning
Supervisor: James Magnuson
Defense committee: Effie Kapnoula, Svetlana Pinet, Drew Mc Laughlin

2017-2022

BSC BIOLOGY
University of Alicante, Spain
Thesis: A review of Computational Modelling of Brain Networks Involved in Language Use and Acquisition (9,2/10)

2021 – 2022

ERASMUS SEMESTER
University of Skövde, Sweden
Coursework in Cognitive Neuroscience

2017 – 2020

PROFESSIONAL CLARINET STUDIES
Jose Iturbi Music Conservatory

EXPERIENCE

Dec 2023 – Present
(full-time)

***RESEARCH ASSISTANT*, SPOKEN LANGUAGE GROUP**

Basque Center on Cognition Brain and Language, Spain

Projects: Connecting Words: The Role of Co-Activation in Lexical Acquisition (CO-LEXI) & The role of familiarity in multi-talker and multi-accent switching costs: A pupillometry investigation (FamPup)
Supervisors: Effie Kapnoula & Drew Mc Laughlin

- Investigate effects of prior knowledge on word learning and whether and how these effects may be modulated by interleaving training.
- Study familiarity effects in multi-talker and multi-accent switching costs.
- Coordinate participant recruitment and scheduling.
- Set up, programming and running of behavioral and psychophysiological experiments (e.g., EEG, eye-tracking, pupillometry).
- Data analysis in Python (e.g., Time-frequency evolution, FFT, Hilbert transform, real-time classifiers, etc.).

Jul 2023 – Present
(part-time)

***RESEARCH COLLABORATOR AND DATA ANALYST,* NEURODEVELOPMENTAL ASSESSMENT & IMAGING LAB**

University of California, San Francisco (US)

Supervisor: Dr. Carly Demopoulos

- Study fine motor abilities of autistic participants in late childhood through adolescence and examine associations between those motor skills and expressive verbal abilities.

- Conducted literature review and data analysis in SPSS, presented findings to the team members.
- Write up the results of experiments as a paper (to be published)

Oct 2022 – Jun 2023

MSC THESIS-RESEARCH ASSISTANT, COMPUTATIONAL NEUROSCIENCE LAB

Basque Center on Cognition Brain and Language, Spain

Project: Catastrophic interference in human speech models
Supervisor: James Magnuson

- Implement and train a neural network with Pytorch by extending the CLS framework to the domain of acquiring lexical knowledge in two languages to avoid catastrophic interference.
- Used packages PyTorch, TensorFlow, Scikit-learn, Pandas, Matplotlib
- Wrote a paper on the findings of the study (to be published) presenting the findings at the 2023 Society for the Neurobiology of Language Annual Meeting in Marseille, FR

Feb 2022 – Jun 2022

RESEARCH INTERNSHIP

Max Plank Institute for Psycholinguistics, Nijmegen, Netherlands

Project*:* A heterogeneous model of word recognition (HetRec)
Supervisors: Prof. P. Hagoort & Alessio Quaressima

- Help implement a biologically informed feedforward network model of spoken word recognition with Python.
- Created custom scripts in python to test standard word recognition experiments.
- Interpret data, analyze results, and produce reports

Feb 2017 – Jun 2021

MANAGER A

Mercadona S.A., Valencia, Spain

- Customer service, cashier, impulse sales, and prescriptions.
- Section and stock management with a produce stock of 50k€ per month
- New employee training, offering continuous advice and mentorship on duties and best practices

AWARDS & SCHOLARSHIPS

Ministry of Education MSc Scholarship, University of Basque Country
Award: 6,000€ plus tuition waiver during the period of 1/9/2022 – 31/6/2023

Ministry of Education BSc Scholarship, University of Alicante
Award: 6,000€ plus tuition waiver during the period of 1/9/2017 – 31/8/2022

Erasmus+ Scholarship for a Traineeship at Max Plank Institute for Psycholinguistics, Nijmegen, Netherlands
Award: 3,000€ during the period of 1/2/2022 – 31/6/2022

Erasmus+ Scholarship for an Exchange Semester at University of Skövde, Sweden
Award: 4,000€ during the period of 1/8/2021 – 31/1/2022

PUBLICATIONS

Marian Simarro González, Thomas Hannagan, Nils Beck, Manuel Carreiras, James Magnuson (In prep). Mitigating catastrophic interference in neural network models of bilingual lexical acquisition.

Marian Simarro González, Gessica Ni, Carly Demopoulos (In prep). Fine motor abilities of autistics in late childhood through adolescence correlate with expressive verbal abilities.

POSTER PRESENTATIONS

James Magnuson, **Marian Simarro González***, Thomas Hannagan, Nils Beck, Manuel Carreiras (2023). Mitigating catastrophic interference in neural network models of bilingual lexical acquisition. Poster presented at the Society for the Neurobiology of Language Conference, Marseille, France (* presenting author)

VOLUNTEERING

2022 – 2023 Gautena (Gipuzkoa Society for Autism)
Autistic Assembly Valencia (Co-founder)

LEADERSHIP

2022 – 2023 Student Council UPV/EHU, Master student's representative
MSc Cognitive Neuroscience of Language UPV/EHU, Class Delegate

COURSES

Jan 2023 **Certified Peer Reviewer** (Elsevier)

Jan 2023 **Scientific Computing with Python** (Free Code Camp)

Jan 2023 **Data Analysis with Python** (Free Code Camp)

Oct 2022 **Natural Language Processing with vector spaces** (DeepLearning.AI, Coursera)

May 2022 **Python for Cognitive Science research** (International Max Planck Research School)
Covered: EEG/ fMRI/ neural signal process and analysis
Packages: NumPy, SciPy, Pandas, Seaborn, MNE, Nilearn

Neurolinguistics (Saint Petersburg State University, Coursera)

SKILLS

Data Analysis	R, MATLAB, SPSS
Programming	Python (numpy, pandas, scikit-learn, pytorch)
Experiment/stimuli software	PsychoPy, PRAAT
Other Software	Excel, Word, Power Point, Canva
Statistics	Regression (basic, hierarchical, generalized), growth curve analysis, generalized additive mixed-effects modeling, Bayesian statistics
Methods experience	Behavioral, pupillometry, eye-tracking, EEG, fMRI, online
Languages	Spanish (native), English (C1), Catalan (C1) French (B2), Italian (B1)